

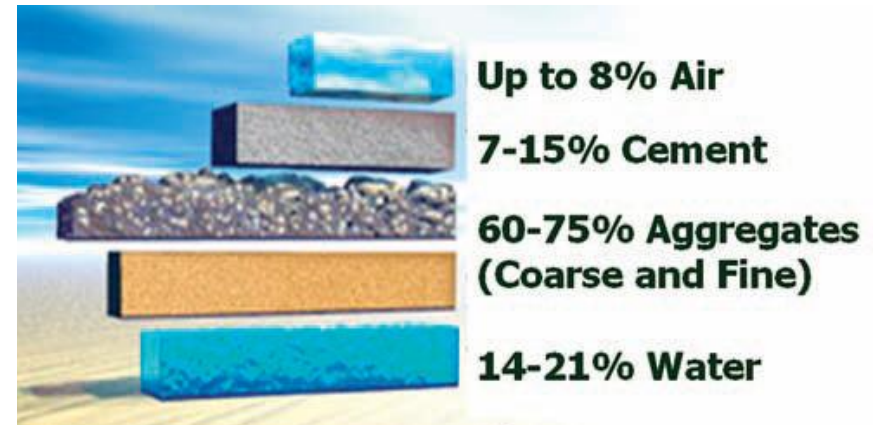
Evaluation of concrete strength

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FACTORS INFLUENCING CONCRETE PROPERTIES

- w/c Ratio
- Cement Content
- Aggregate
 - Maximum aggregate size
 - Aggregate grading
 - Aggregate shape and texture
 - Sand/coarse aggregate ratio
 - Aggregate/cement ratio
- Admixtures
- Mixing Procedures
- Curing
- Age



Water

- Water < 2000 ppm of total dissolved solids is satisfactory for making concrete.
- Water > 2000 ppm of dissolved solids should be tested for its effects on strength and time of set.

Parameter	Cl ⁻ content	SO ⁴⁻ content	Solid material content	Comparative samples strength
prEN 1008	≤ 600 mg/L (concrete/grout) ≤ 1200 mg/l (reinforced concrete)	≤ 2000 mg/L	≤ 1% w.t.	The mean 7-days and 28-days compressive strength of the mortar or concrete samples wash water must be at least 90% of the mean strength of the control samples (prepared with distilled or tap water)
ASTM C94	≤ 400 mg/l (prestressed concrete) ≤ 1200 mg/l (reinforced concrete)	≤ 3000 mg/l (Optional)	≤ 50000 mg/l	The mean 7-days and 28-days strength of the concrete samples prepared with prepared with wash water must be at least 90% of the mean strength of the control samples (prepared with distilled or city water)

Acceptance Criteria for Questionable Water

	Limits	Test method
Compressive strength, minimum percentage of control at 7 days	90	ASTM C31, C39
Time of set, deviation from control, hr:min.	from 1:00 earlier to 1:30 later	ASTM C403

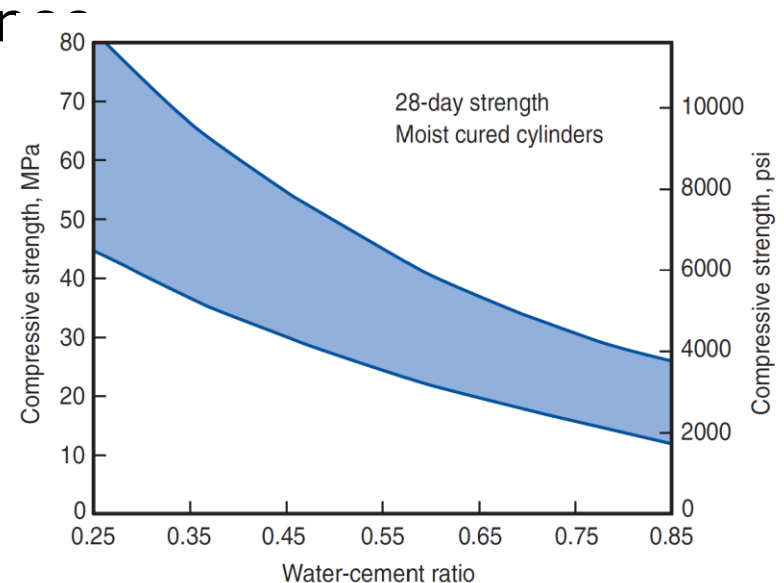
* Comparisons must be based on fixed proportions of a concrete mix design representative of questionable water supply and a control mix using 100% potable water.

w/c Ratio Parameters

- Aggregate size
- Grading of Aggregate
- Surface texture of aggregate
- Shape of aggregate
- Cement type and source
- Pozzolans
- Air Entraining & Chemical Admixtures
- Setting Time

Water in Concrete

- Increased water:
 - reduced strength
 - increased shrinkage and creep
 - increases permeability
 - reduced abrasion resistance
 - reduced Freeze-Thaw resistance



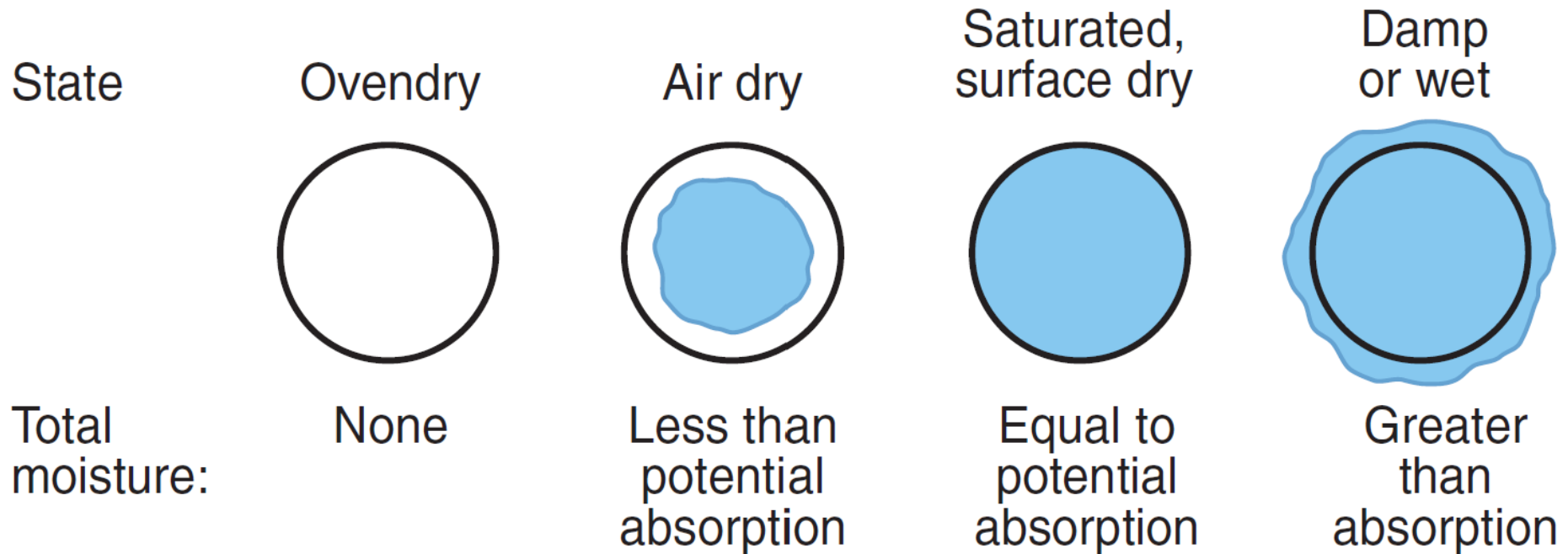
Aggregate in Concrete

- Aggregate is the second most influential ingredient in concrete.
- Aggregate
 - Occupies 60-75 % of volume
 - Fine Aggregate is typically 35-45 % of total aggregate
 - Mortar (Air, water, cement, fly ash, sand) is typically 50 - 65 % of total volume of a mixture

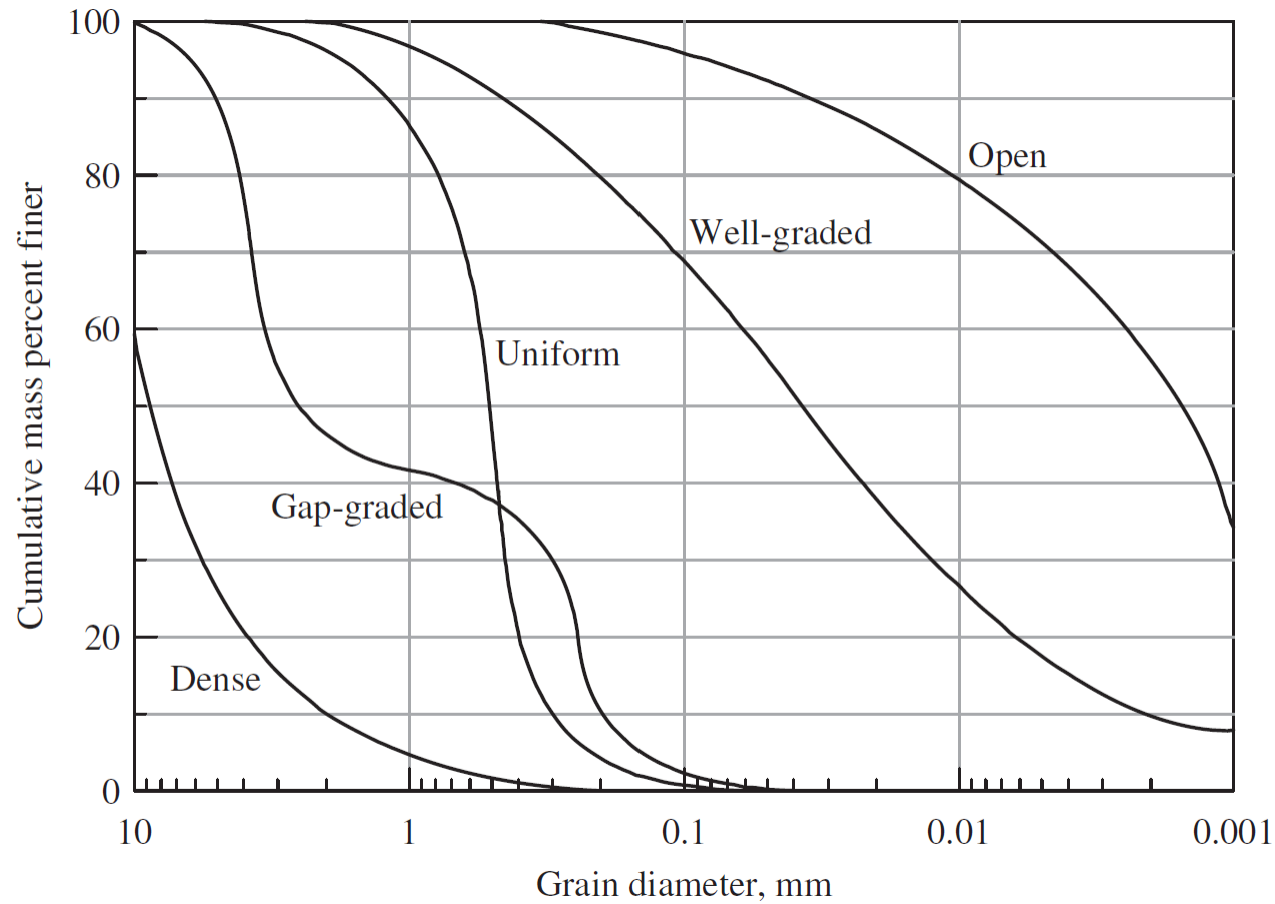
Aggregates size

- Fine: Sand or Crushed Stone ($< 5\text{mm}$)
- Coarse: Gravel or Crushed Stone (5-50 mm)
- Aggregate must be washed in many areas
 - Granite & other crushed stone
 - Recycled concrete
- All must satisfy ASTM C33

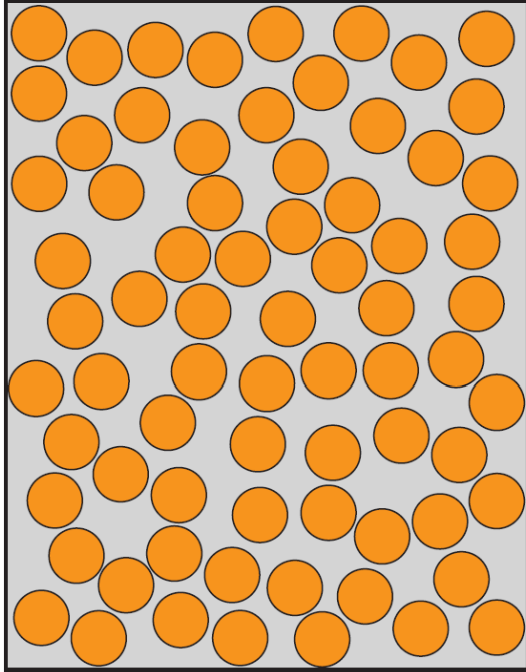
Aggregate moisture



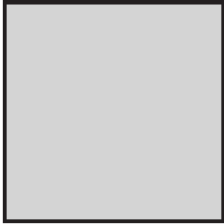
Grading Aggregates



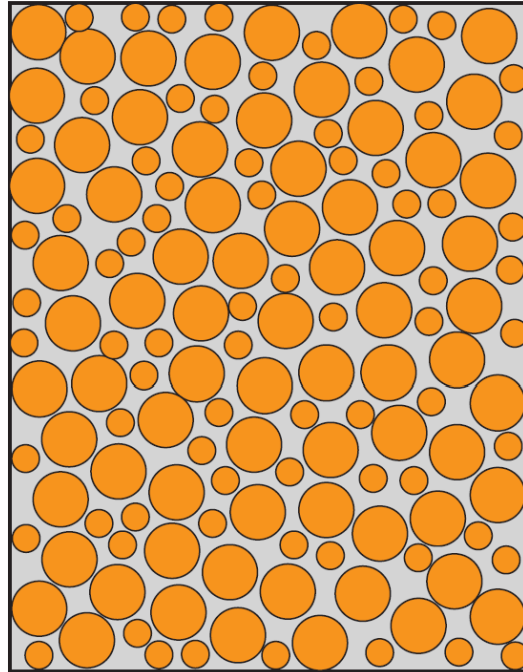
Single-sized



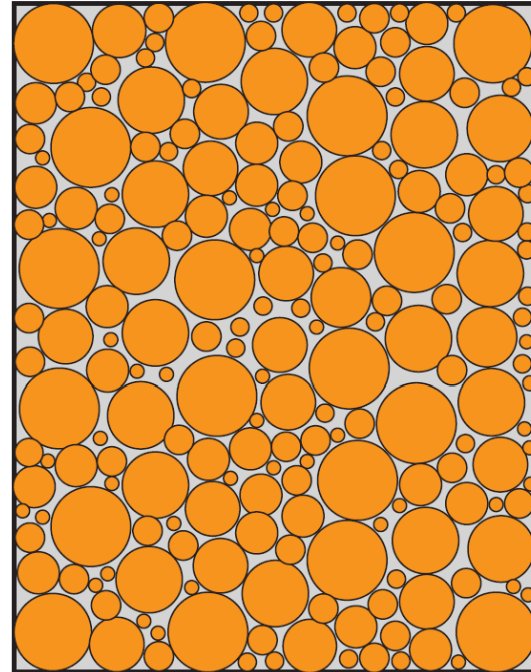
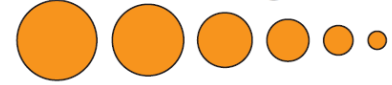
Resulting
paste content



Poorly-graded



Well-graded

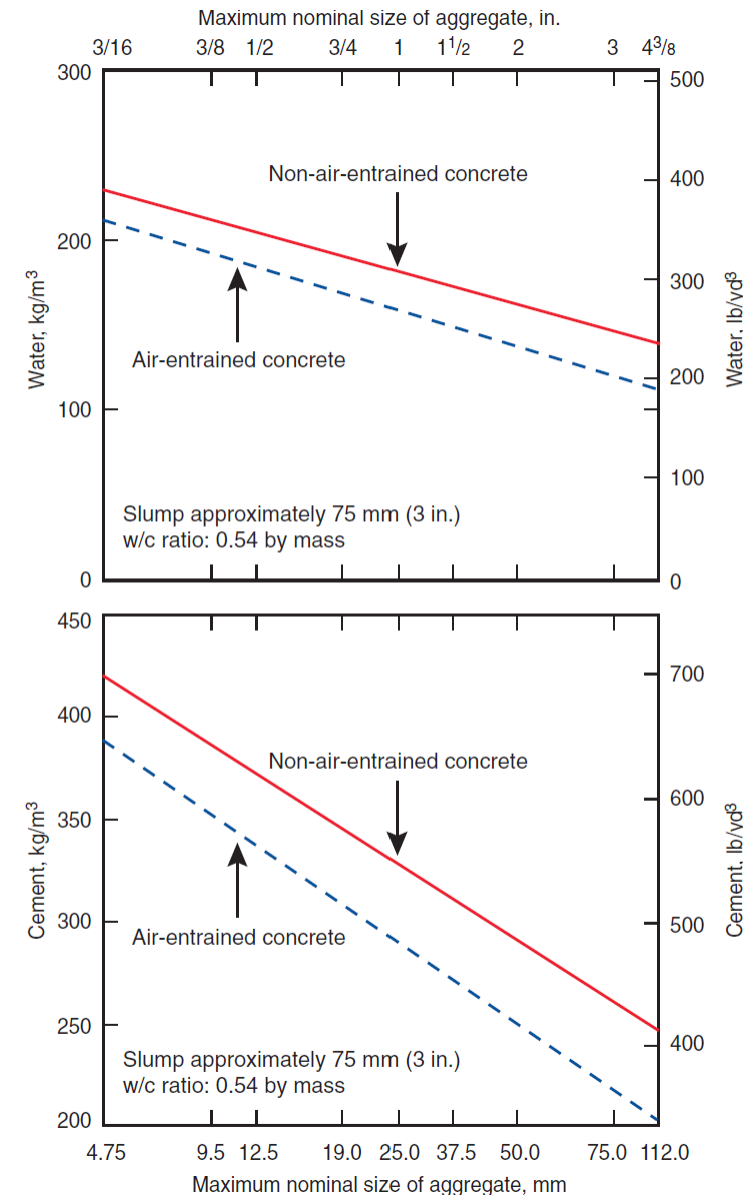


Absolute volume of concrete

Influence of Aggregates

- Source
- Shape
- Surface texture
- Size
- Grading

Affected strength	Relative effect (%) of	
	Shape	Surface texture
Compressive	22	44
Flexural	31	26



EFFECT OF CHANGING FINENESS MODULUS ON CONCRETE PROPERTIES



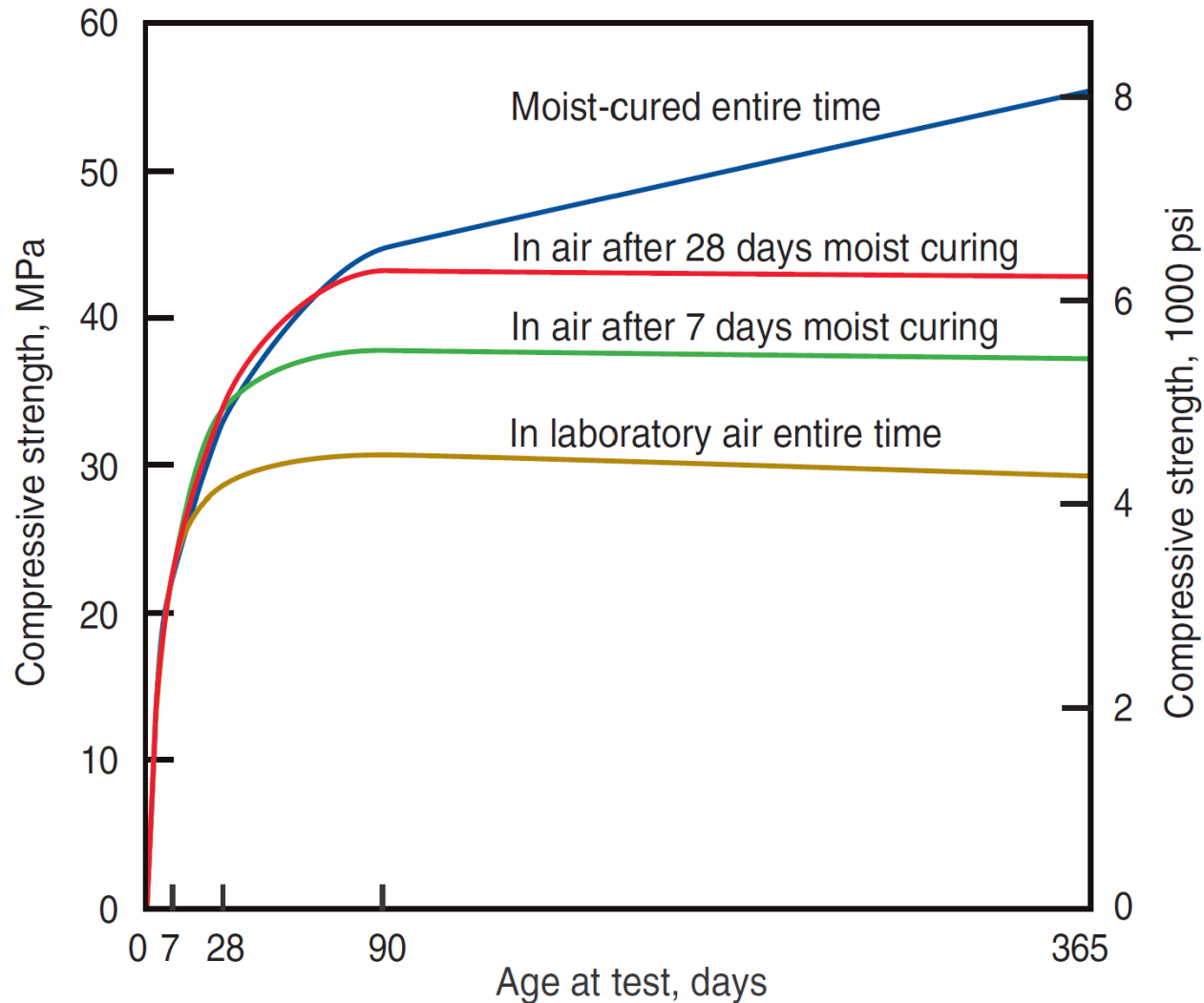
CONCRETE PROPERTY	DECREASING FM (FINE SAND)	INCREASING FM (COARSE SAND)
Water Requirements	MORE	LESS
Water-Cement Ratio	HIGHER	LOWER
Strength	LOWER	HIGHER
Finishability	EASY	DIFFICULT

Note: Fineness Modulus: Sum of Cumulative % Retained/100. The FM should range between 2.3 and 3.1, and not vary more than 0.2 from the typical value of the aggregate source.

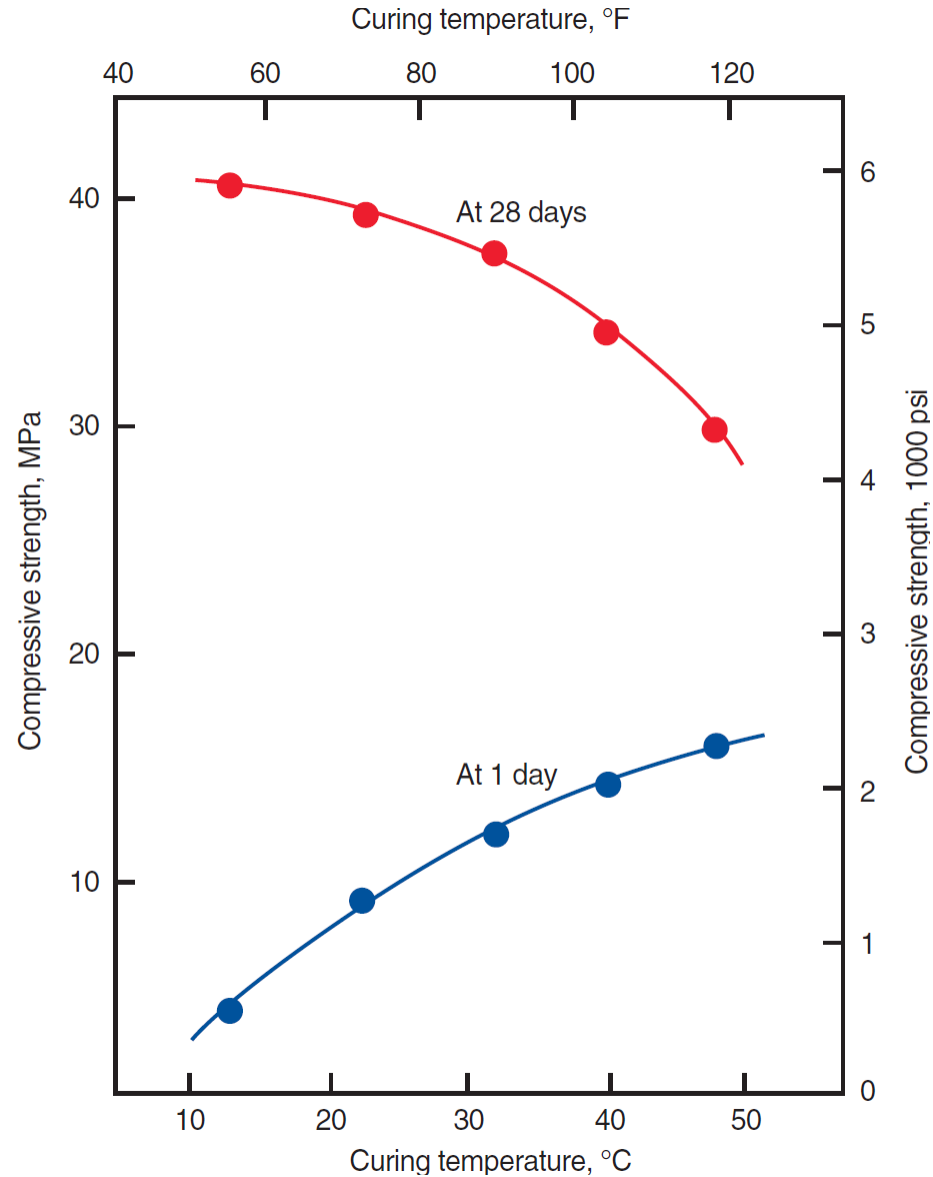
Choosing Aggregate Size

- maximum nominal size of aggregate
 - $\frac{1}{5}$ smallest dimension
 - $\frac{1}{3}$ thickness of slab
 - $\frac{3}{4}$ clearance between rebars
- Congestion
- Shrinkage
- Mass Concrete

Curing Concrete



Curing Concrete



Concrete - Hardened Properties

- Strength Gain
 - Normal strength concrete 7-day f_c is 60-70% of the 28-day for Type I
 - 3-day f_c is about 50% of the 7 day.
 - Type III may have 3-day f_c of 60-70% of the 28-day
 - Moist cured concrete gains faster than air dried
 - Steam curing is fastest

Concrete Strength

- Tensile Strength: tensile strength can be estimated by
 - $\approx 7.5 \sqrt{f_c'}$
 - 10% of compressive strength
- Flexural Strength: Measure of cracking strength.
 - (pavement and slabs on grade applications)
 - Flexural Strength is generally $7.5 - 10 \sqrt{f_c'}$
- Shear Strength 20% of compressive strength

Strength Test-Cube specimen (BS EN 12390-3: 2019)

- BS EN 12390 allows the use of a cube, cylinder, or core (from field concrete)
- The most frequently used standard size of cube specimens is $150 \times 150 \times 150$ mm
- The nominal size of specimen should be chosen to be at least 3.5 times the maximum aggregate size
- A cube specimen should be prepared by pouring concrete into a cubic mold in at least two layers, but no layer should be thicker than 100 mm.
- Each layer should be stroked at least 25 times by a hemispherical-tipped steel rod (16mm diameter) or a steel compacting bar having a square (25mm $\times$ 25 mm) cross-section.
- The compacted specimens need to be kept in the mold for at least 16 hours (but no longer than 3 days), protected (often by covering) against dehydration at $20 \pm 5^\circ\text{C}$.
- After demolding, the specimen should be cured at a temperature of $20 \pm 1^\circ\text{C}$ in water or in a chamber at a relative humidity of no less than 95%.

Strength Test-Cylinder specimen (ASTM C39/C39M)

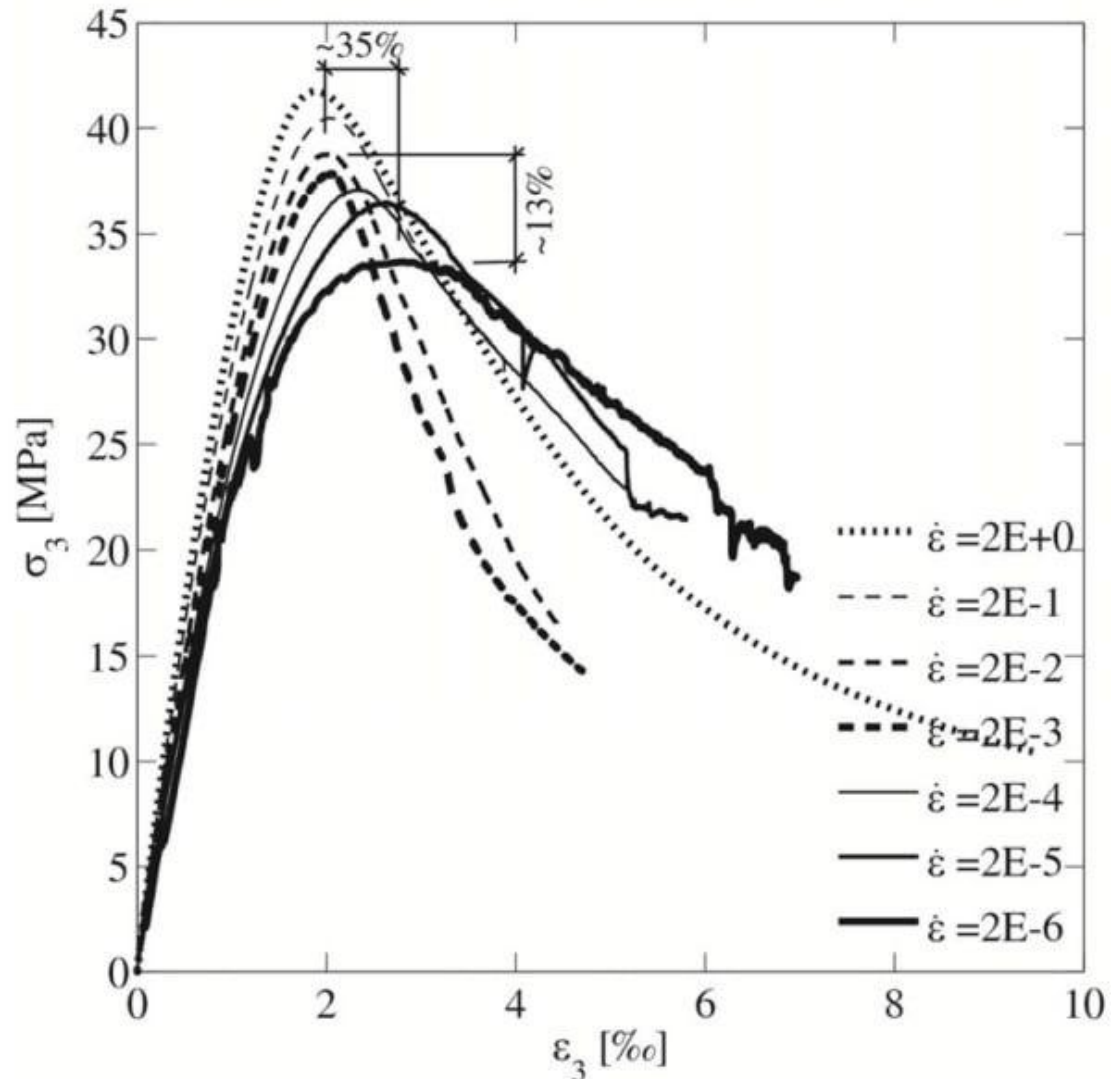
- ASTM C39 allows the use of cylindrical specimens of various sizes (length/diameter = 2; and diameter shall be at least 3 times the nominal maximum aggregate size).
- The most widely adopted standard cylinder size for compression testing is 150×300 mm.
- It should be prepared by pouring concrete into the cylindrical mold in three approximately equal layers, with each layer being stroked 25 times by a hemispherical-tipped steel rod (16 ± 2 mm).
- After 16–24 hours from casting, the specimens should be demolded and cured at a temperature of $23.0 \pm 2.0^{\circ}\text{C}$ in a water tank or moist conditions (ASTM C192/C192M).
- the upper surface of the cylinder is never smooth; hence, grinding or capping is needed to level and smooth the compression surface before a test.

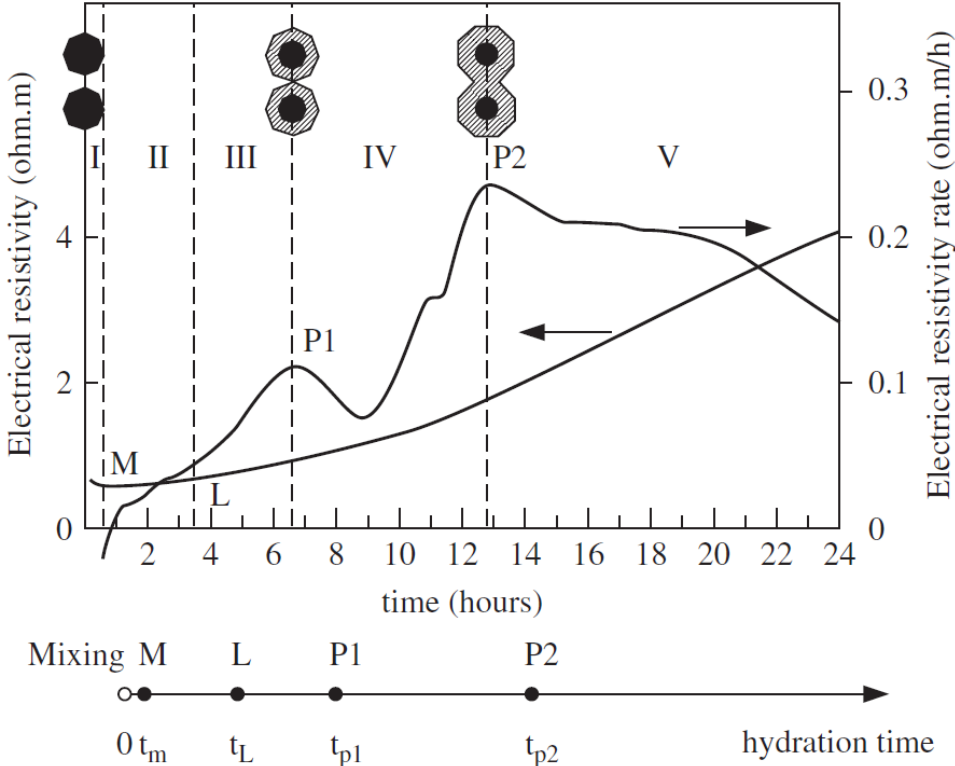
Strength Test

- Loading rate
- End condition
- Size effect

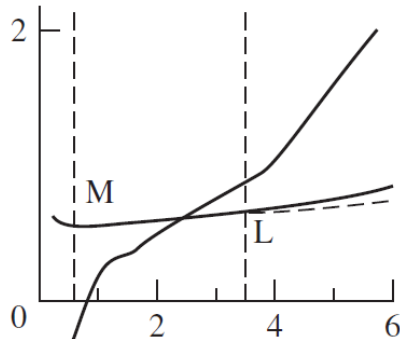
ASTM: $0.25 \pm 0.05 \text{ MPa/sec}$

BS: $0.2\text{--}0.4 \text{ MPa/sec}$



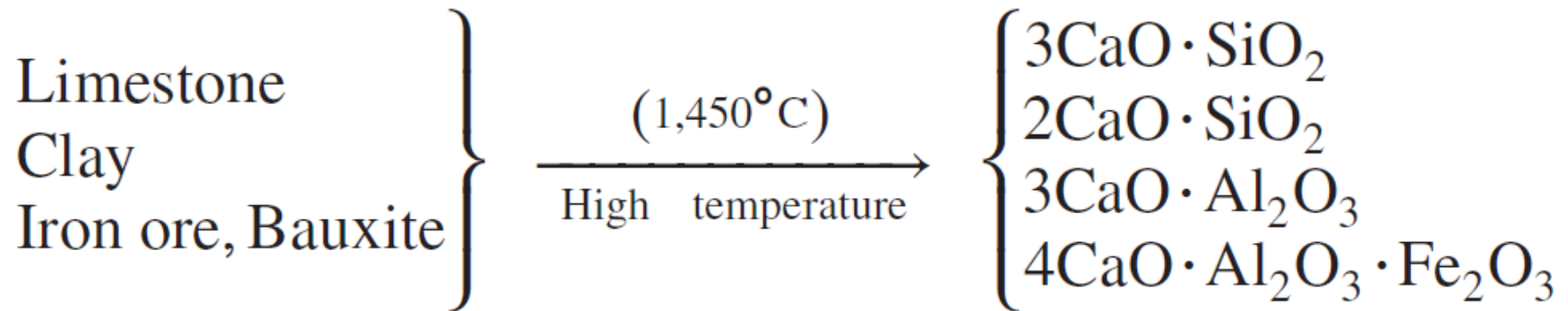


Electrical resistivity response				Setting time (ASTM C191)	
t_m	t_L	t_{p1}	t_{p2}	initial t_{ini}	final t_{fin}
0.73h	3.5h	6.7h	12.8h	3.75h	6.28h



Electrical resistivity development and the rate in resistivity of the cement paste during the first 24 h

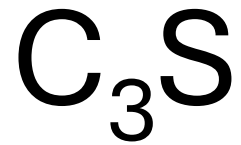
Sources for Cement



- Lime (CaO)
 - Limestone, shale
- Silica (SiO_2)
 - Clay, sand, shale
- Alumina (Al_2O_3)
 - Clay, fly ash, shale
- Iron (Fe_2O_3)
 - Clay, iron ore

Portland Cement Production

Oxide	Shorthand notation	Common name	Weight percent
CaO	C	lime	64.67
SiO ₂	S	silica	21.03
Al ₂ O ₃	A	alumina	6.16
Fe ₂ O ₃	F	ferric oxide	2.58
MgO	M	magnesia	2.62
K ₂ O	K	alkalis	0.61
Na ₂ O	N	alkalis	0.34
SO ₃	$\bar{S}$	sulfur trioxide	2.03
CO ₂	$\bar{C}$	carbon dioxide	—
H ₂ O	H	water	—



Tri Calcium Silicate

- $3\text{CaO}.\text{SiO}_2$
- Provides Early strength development
- 70% reacts by 28 days
- Less resistance to sulfate

C_2S Dicalcium Silicate

- $2CaO.SiO_2$
- Provides late strength development
- 30% reacts by 28 days
- Provides less heat in hydration

C_3A Tricalcium Aluminate

- $3CaO \cdot Al_2O_3$ - “Aluminate”
- Provides Early strength development
- Provides higher heat in hydration
- Low resistant to sulfate attack

C_4AF Tetracalcium Aluminoferrite

- $4CaO \cdot Al_2O_3 \cdot Fe_2O_3$ - “Ferrite”
- Iron facilitates formation of other compounds-acts as a flux
- Little contribution to strength

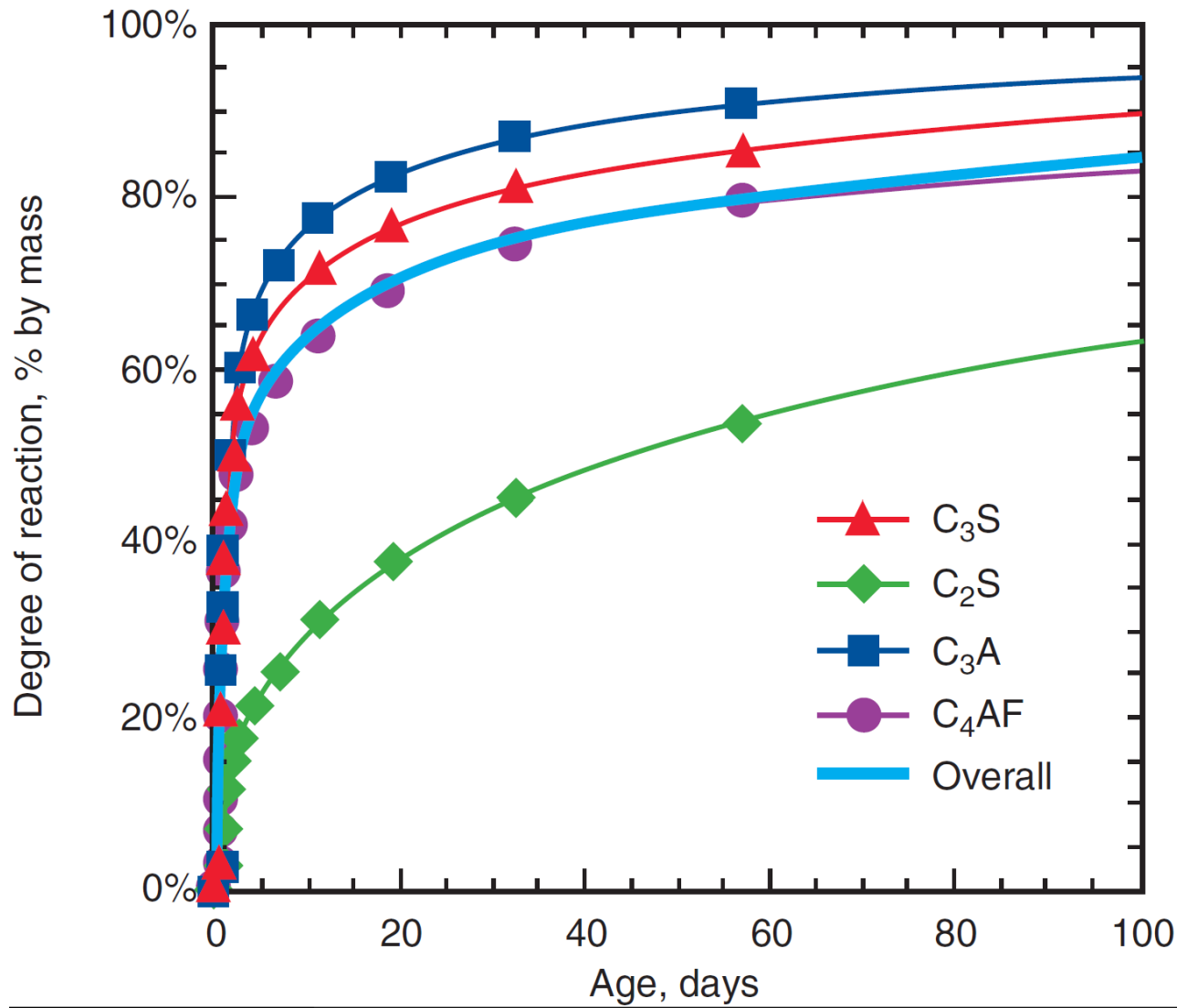
Hydration

- C_3S and C_2S = ~ 75% of the weight of Portland Cement
- React with Water to form two new compounds:
 - **Calcium Hydroxide**
 - **Calcium Silicate Hydrate (CSH)**

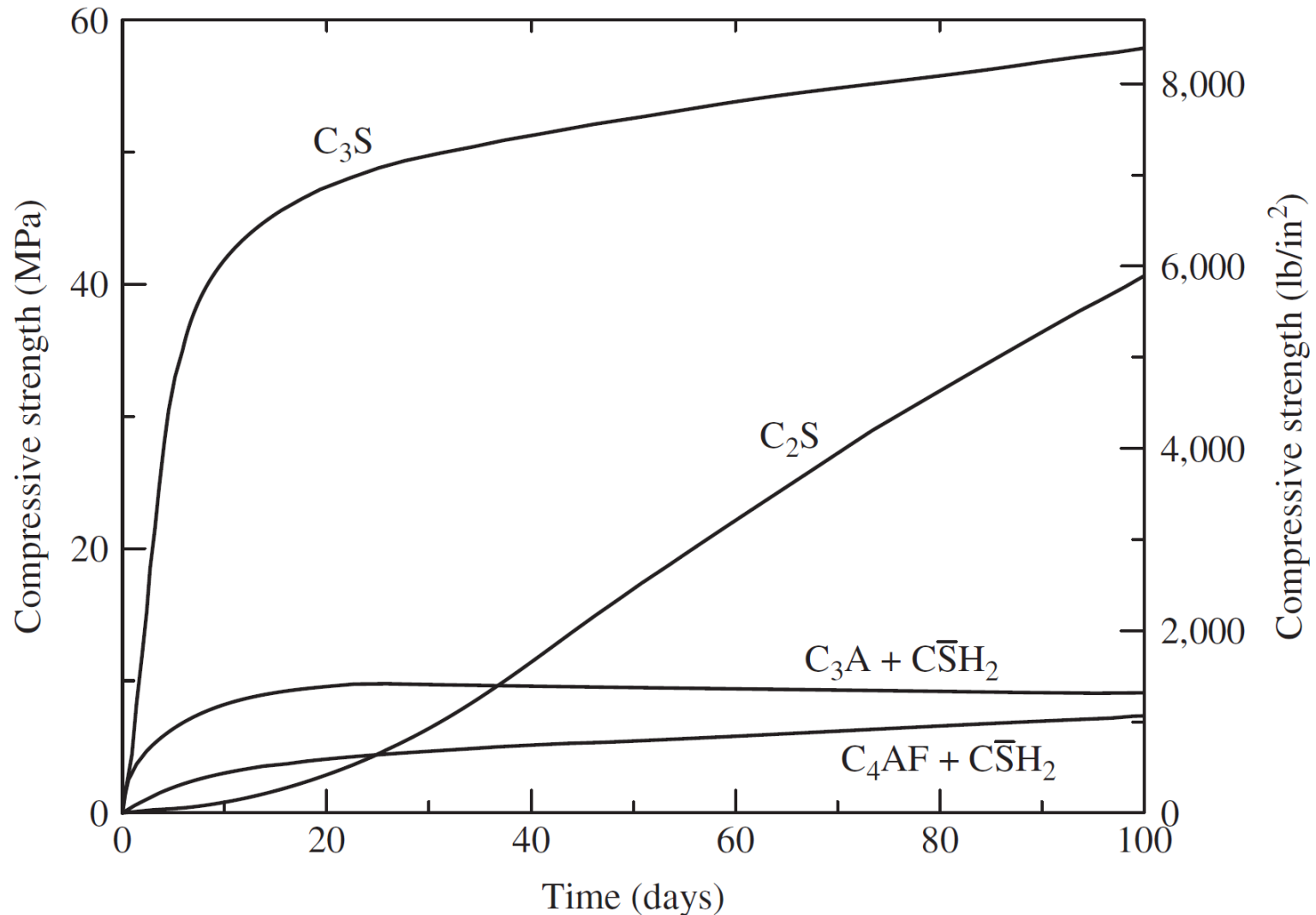
- Hydration:



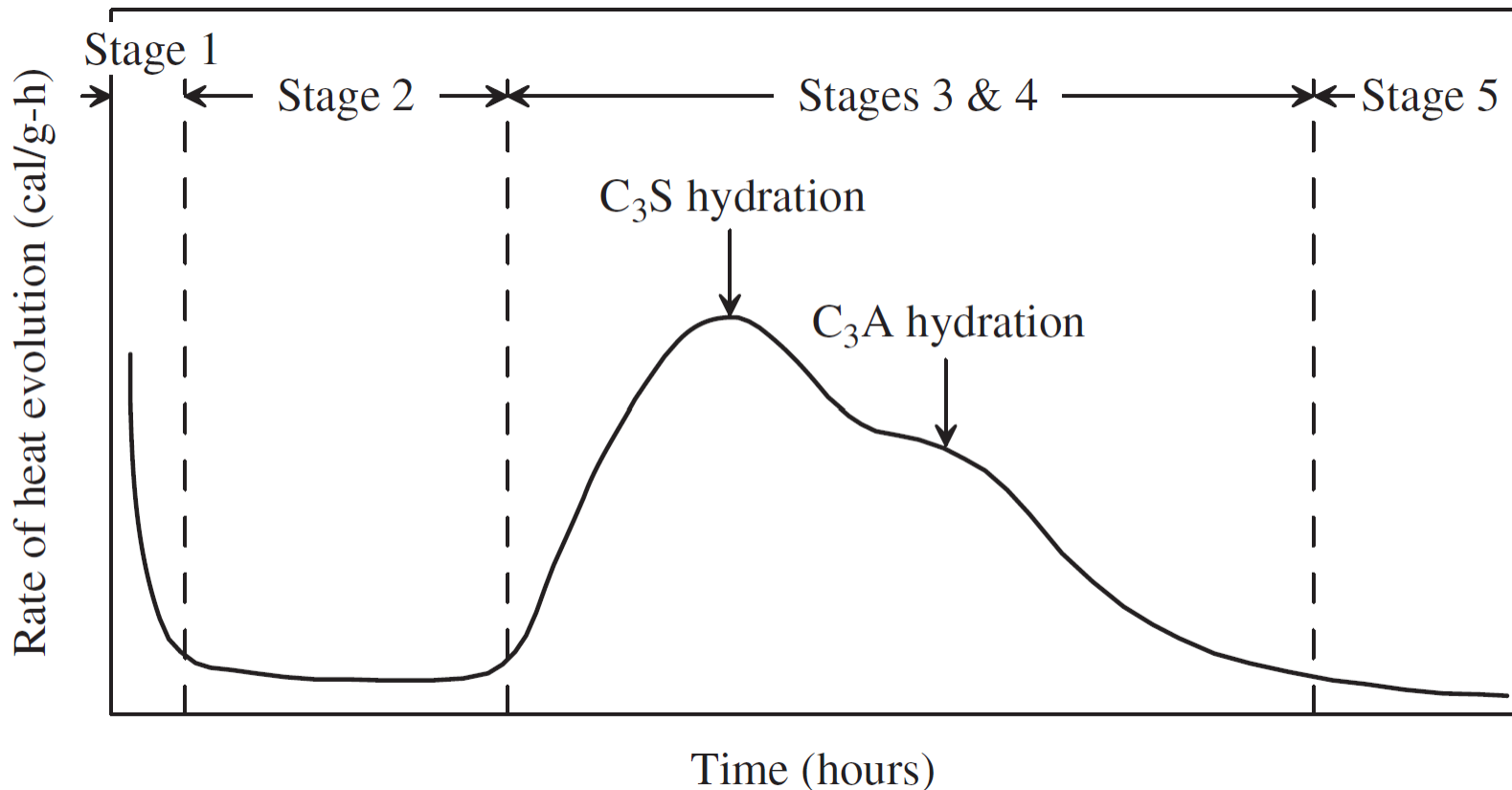
Hydration process of primary constituents of Portland cement



Strength development of primary constituents of Portland cement



A typical calorimetric curve of Portland cement

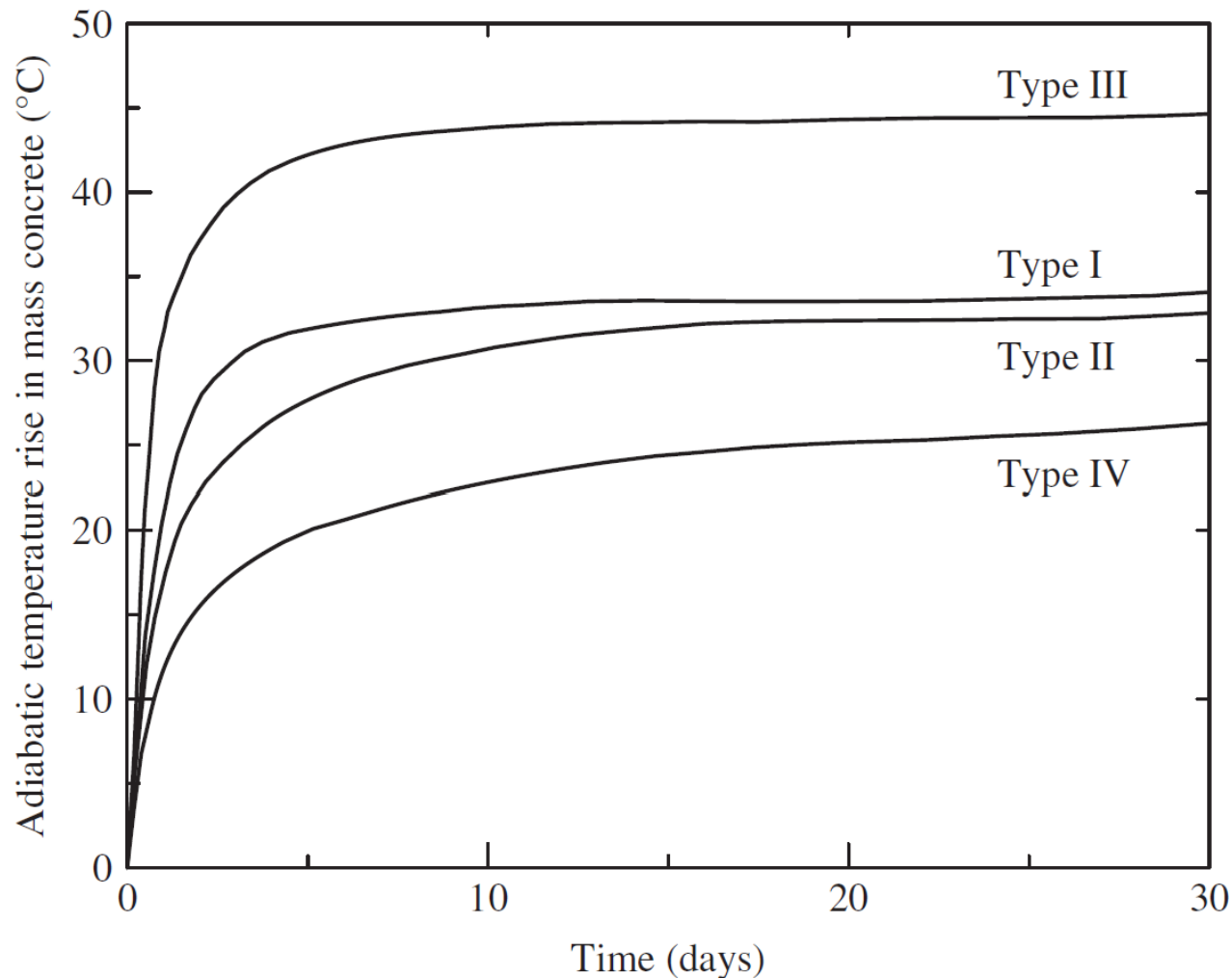


Chemical compositions and physical properties of different Portland cements

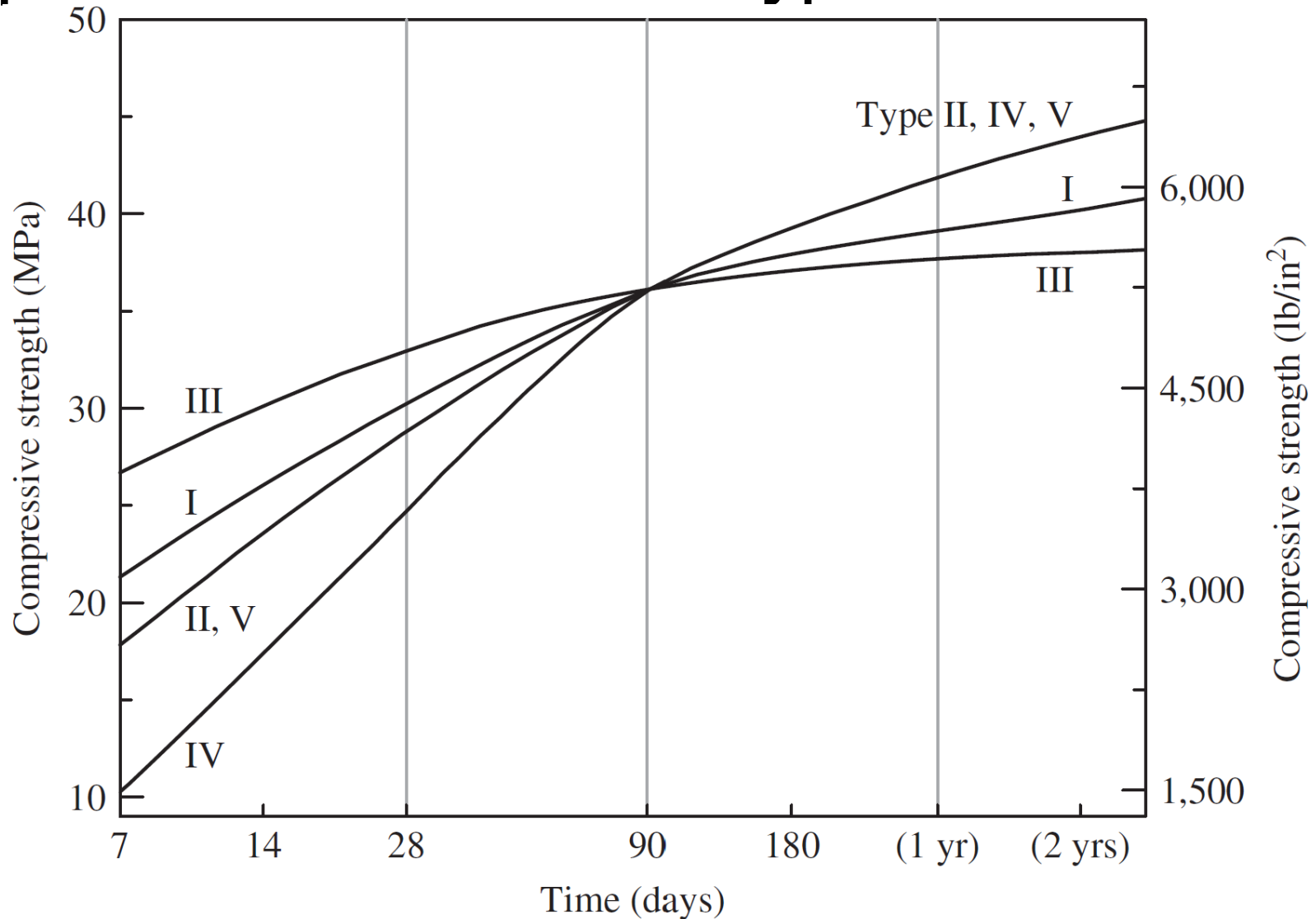
- Type I Normal*
- Type II Moderate Heat and Sulfate*
- Type III High-Early Strength*
- Type IV Low Heat of Hydration
- Type V High Sulfate Resistance

Chemical compositions and physical properties	Portland cement type				
	I	II	III	IV	V
C_3S	50	45	60	25	40
C_2S	25	30	15	50	40
C_3A	12	7	10	5	4
C_4AF	8	12	8	12	10
CSH_2	5	5	5	4	4
Fineness (Blaine, m^2/kg)	350	350	450	300	350
Compressive strength (1-day, MPa [psi])	7 [1,000]	6 [900]	14 [2,000]	3 [450]	6 [900]
Heat of hydration (7-day, J/g)	330	250	500	210	250

Adiabatic temperature rise in mass concretes with different types of cement



Strength development of cement pastes with different types of cement



TYPES OF CONCRETE

Classification	Unit weight (kg/m ³)
Ultra-lightweight concrete	$< 1,200$
Lightweight concrete	$1,200 < UW < 1,800$
Normal-weight concrete	$\approx 2,400$
Heavyweight concrete	$> 3,200$

Classification	Compressive strength (MPa)
Low-strength concrete	< 20
Moderate-strength concrete	$20 \sim 50$
High-strength concrete	$50 \sim 150$
Ultra high-strength concrete	> 150

Advantages of Concrete

- *Economical*
- *Ambient temperature-hardened material*
- *Ability to be cast*
- *Energy-efficient*
plain concrete: 450–750 kWh/to structural steel: 8000 kWh/ton
- *Excellent water resistance*
- *High-temperature resistance*
- *Ability to consume waste*
- *Ability to work with reinforcing steel*
- *Less maintenance required*

Limitations of Concrete

- *Quasi-brittle failure mode*
- *Low tensile strength*
- *Low toughness (ductility)*
- *Low specific strength (strength/density ratio)*
- *Formwork is needed*
- *Long curing time*
- *Working with cracks*

Concrete - Fresh Properties

- Workability: Ease with which a concrete can be handled and placed into forms.
 - Slump
 - Kelly Ball
 - Penetration
 - Flow Cone

Concrete - Workability

- cement: too fine of material
 - stickiness
 - increased water demand
- water: too much water
 - segregation
 - bleeding
- water: too little water
 - harshness
 - compaction problems
- fly ash: increases flow
 - ball bearing effect
 - ionic effect
 - reduced water demand
- aggregate
 - rounded particles flow more easily
 - Too much sand
“stickiness”
 - Poor gradation - harsh

Concrete - Fresh Properties

- Pumpability: Ease with which a given mix can be pumped without segregation or loss of properties
 - aggregate: rounded particles pump more easily
 - water: too much - segregation, too little - friction
 - cement: too little - blow through,
 - fly ash: helps prevent segregation, better flow

Concrete - Fresh Properties

- Compactability: Ease with which a given mix can be fully compacted to eliminate the trapped air.
 - harshness
 - gradation
- Finishability: Ease with which a given mix can be fully finished with the desired texture
 - stickiness
 - harshness

Concrete - Fresh Properties

- Setting Time
 - Cement: different cements have different setting times
 - alkalis, sugars, salts, organics
 - Water: Impurities
 - -sodium carbonate (Na^+) rapid set
 - -bicarbonate can accelerate or retard set
 - Aggregate: None

Concrete - Fresh Properties

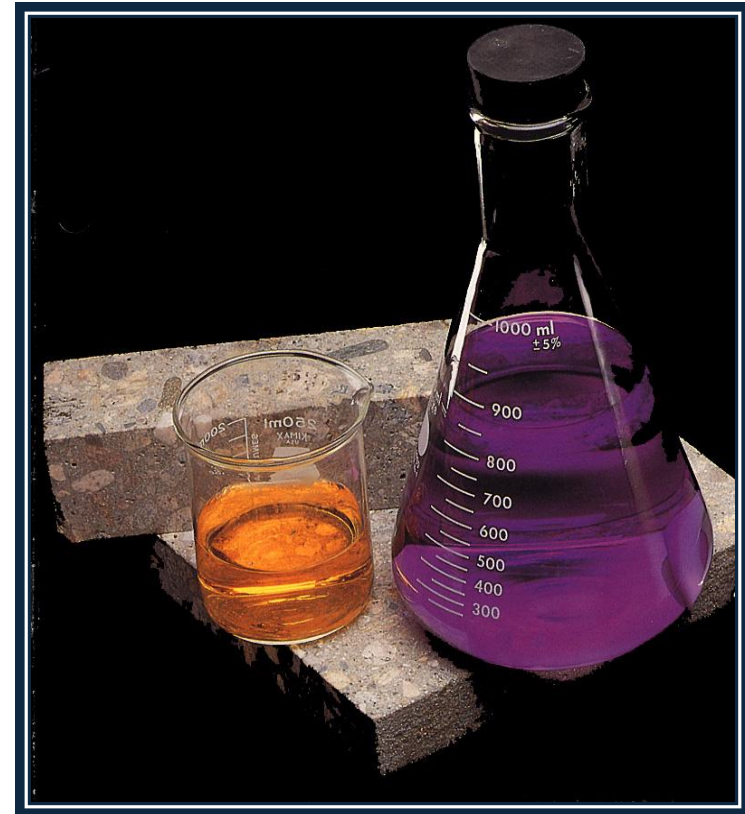
- Bleeding: rate of surface water exceeds the evaporation rate.
 - Water: too much water (severe bleeding), too little water (surface drying)
- Air Content
 - Water: -too much increases entrapped air voids
 - -too little doesn't disperse Air Entraining Agent properly
- Unit Weight

Admixtures

- DEFINITION:

Admixtures are any ingredients in concrete other than:

- Water
 - Aggregates
 - Cementitious Materials
 - Fiber Reinforcement
- Added to the batch before or during mixing



Why Use Admixtures?

To Modify fresh concrete properties

- decrease water content
- increase workability
- retard or accelerate setting time
- reduce segregation
- reduce the rate of slump loss
- improve pumpability, placeability, finishability
- modify the rate and/or capacity for bleeding

Why Use Admixtures?

To Modify hardened concrete properties

- improve impact and abrasion resistance
- inhibit corrosion of embedded metals
- reduce plastic shrinkage cracking
- reduce long term drying shrinkage
- produce colored concrete
- produce cellular concrete

Current Admixture Standards

(AASHTO Designations in parentheses)

- Air Entraining ASTM C 260 (M 154)
- Chemical ASTM C 494 (M 194)
- Calcium Chloride ASTM D 98 (M 144)
- Foaming Agents ASTM C 869
- Admixtures for shotcrete ASTM C 1141
- Flowing Concrete ASTM C 1017
- Grout Fluidifier ASTM C 937
- Pigments ASTM C 979

Air Entrainment

- DEFINITION: Air-Entraining Agents are primarily used to stabilize tiny bubbles generated in concrete to protect against freezing and thawing cycles.



Chemical Admixtures

- Dispersing Agents
 - Water Reducers, Superplasticizers
- Accelerators
- Retarders



ASTM C 494 Chemical Admixtures (AASHTO M 194)

- Type A - Water-reducing admixtures
- Type B - Retarding admixtures
- Type C - Accelerating admixtures
- Type D - Water-reducing and retarding
- Type E - Water-reducing and accelerating
- Type F - High range water reducing
- Type G - HRWR and retarding

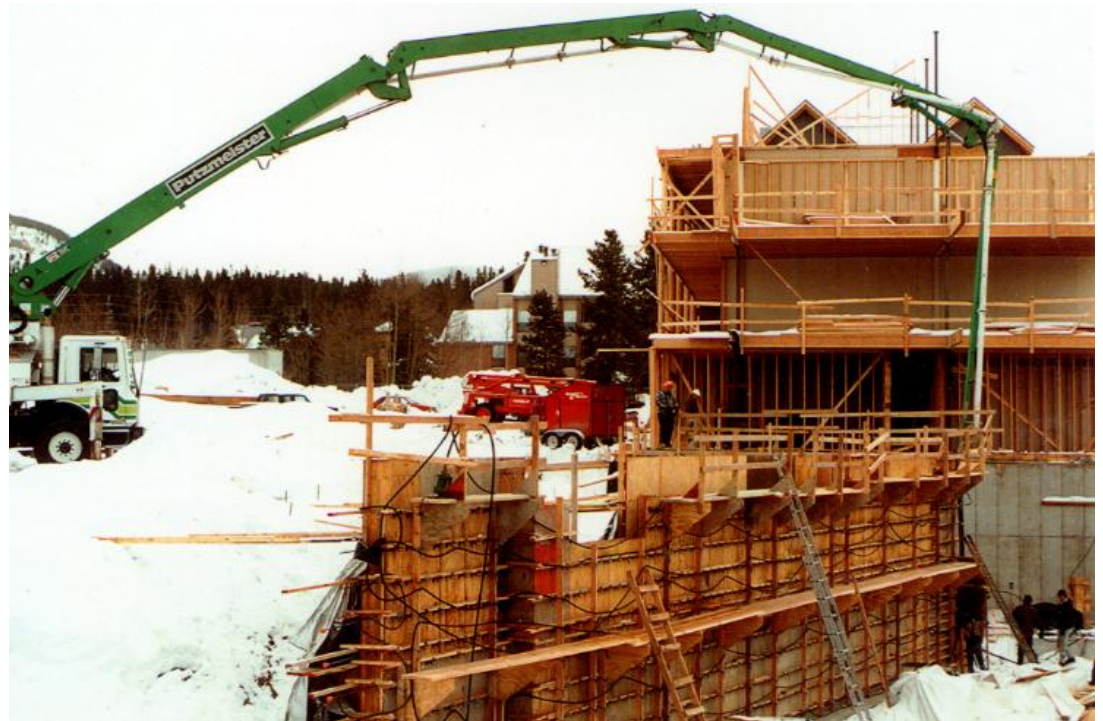
Water Reducers

- DEFINITION: Water Reducers are used for the purpose of reducing the quantity of mixing water required to produce a concrete of given consistency.



Accelerators

- DEFINITION: Accelerating admixtures are added to concrete for the purpose of shortening set time and accelerating early strength development.



Retarders

- DEFINITION: Retarding, and Water-reducing and retarding admixtures are used to offset acceleration and unwanted effects of high temperature and keep concrete workable during placement and consolidation.



Shrinkage Reducing Admixtures

- DEFINITION: Shrinkage Reducing Admixtures are used to minimize drying shrinkage cracking in concrete .



Corrosion Inhibitors

- DEFINITION: Corrosion Inhibitors are used to mitigate corrosion of reinforcing steel in concrete.



ASR Inhibitors

- DEFINITION: ASR Inhibitors (primarily Lithium) are used to mitigate alkali-silica reactivity in concrete.



Specialty Admixtures

- Coloring Admixtures
- Workability Agents
- Bonding Admixtures
- Damp-proofing Admixtures
- Permeability-Reducing
- Grouting
- Gas-forming
- Anti-Washout
- Foaming
- Pumping Aids

The Effectiveness of an Admixture

- Depends on:
- Type & Brand
- Amount of Cement
- Water Content
- Temperature
- Aggregate Shape
- Proportions
- Mixing Time
- Consistency of the Mix
- Sequencing



Supplementary Cementing Materials

- **DEFINITION:** Powdered or pulverized materials added before or during mixing to improve or change some of the plastic or hardened properties of concrete.

- Cementitious
- Pozzolans
- Nominally Inert



Cementitious Materials

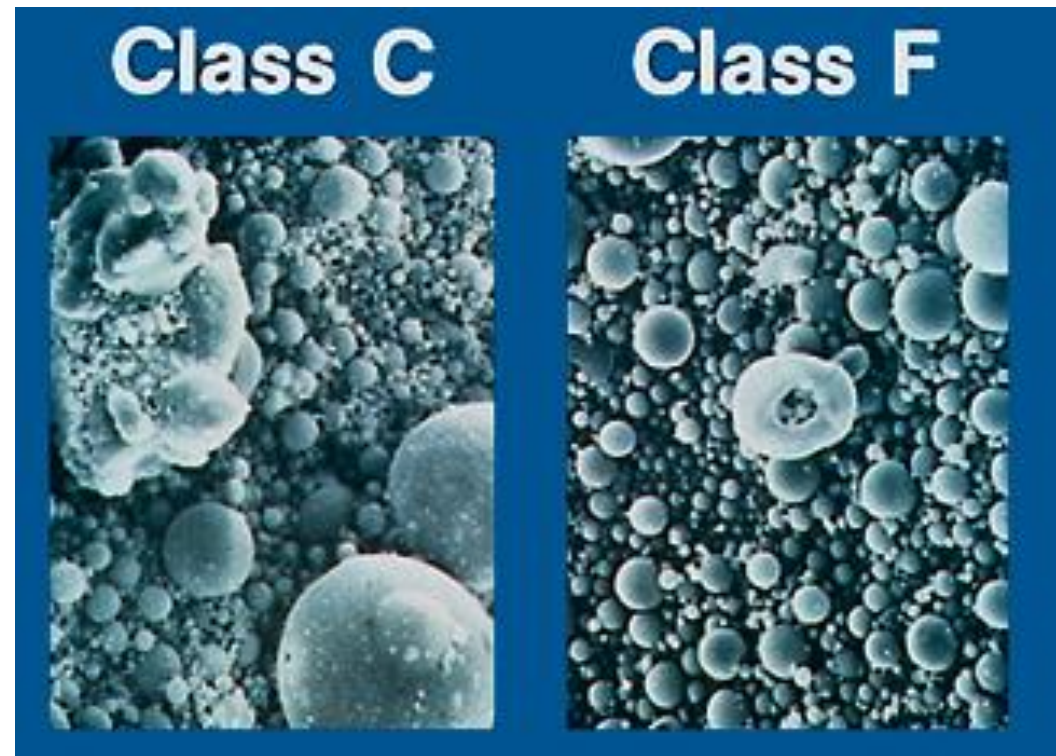
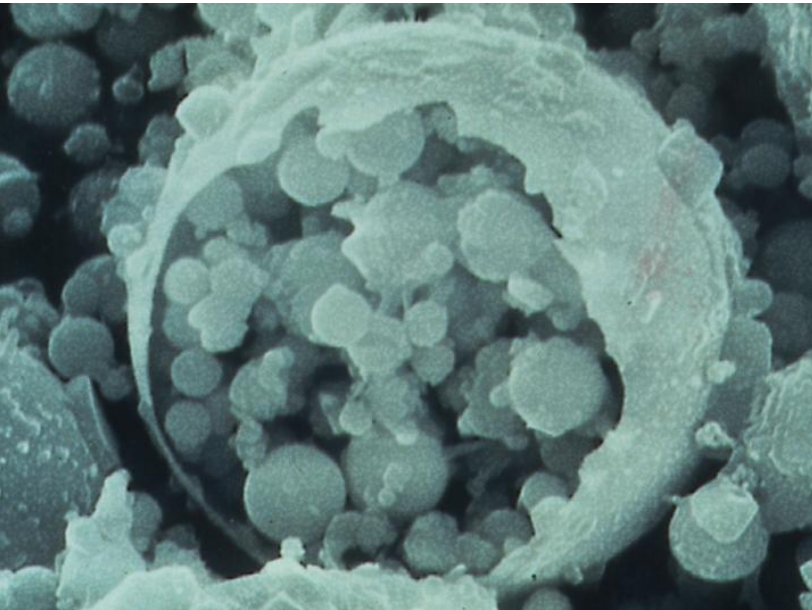
- Possess hydraulic cementing properties
- GGBF slag (by-product of steel industry)
- Natural cement- Cement Rock
- Hydraulic hydrated lime

Pozzolans

- Possess no cementitious value until finely divided and mixed with water and cement
- Cherts, clays, shales
- Fly ash (by-product of coal)
- Silica fume (silicon manufacture)

Fly Ash

- Class F (low calcium) - from burning anthracite or bituminous coal, is pozzolanic
- Class C - from burning sub-bituminous or lignite coal, is somewhat cementitious



GGBFS (Slag)

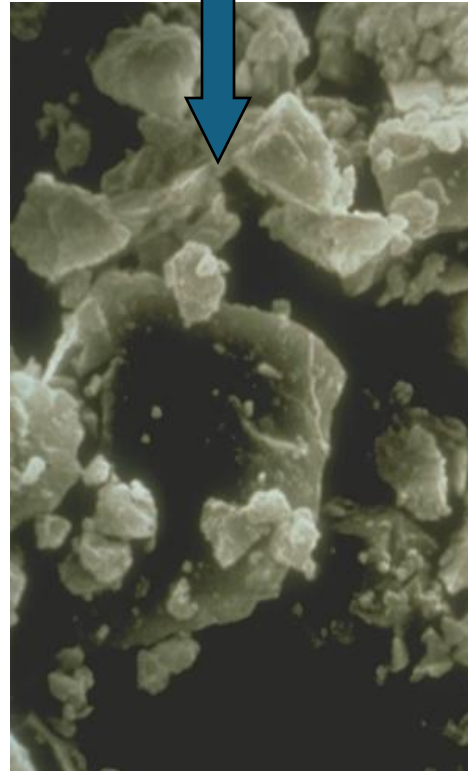
- Formed when molten iron blast furnace slag is rapidly chilled (quenched) by immersion in H_2O
- Grades 80, 100, 120
- Used as a cement replacement



Silica Fume

- Also known as micro-silica
- By-product of the production of silicon and ferrosilicon alloys.
- A small part of silica fume can be used to replace a large part of cement

**Portland
Cement**



**Silica
Fume**



Concrete Mixture Pre-Design

- Engineer
 - Define strength, congestion and durability properties
- Architect
 - Defines color, texture,
- Contractor
 - Defines workability, setting time, ..
- Concrete Supplier
 - Defines aggregates, cement, fly ash, admixtures....

Concrete Mixture Design

- Discussion of defined needs
- Negotiation on conflicting needs
- Negotiation on economics
- Engineer Accepts Mixture Proportions
- Conflicts defined
- Trial Solution determined
 - trial batching
 - trial batching
 - trial batching

Mixture Design Procedures

Step 1: Choose Slump

PCA Table 7-7

- 3” foundations, footings and slabs
- 4” beams, columns & reinforced walls
- 2” mass concrete
- 3” Pavements
- (add 1” for non-vibrated concrete)

Mixture Design Procedures

Step 2: Select Aggregate

- Local Availability
- Large Aggregate reduces water demand
- max size of aggregate
 - $1/5$ minimum size of form dimension
 - $3/4$ minimum rebar spacing
 - $1/3$ the slab thickness

Mixture Design Procedures

Step 3: Choose Air Content

PCA Table 7-6

- *mild* exposure 3-4.5%
non-freezing and non-chemical environment
- *moderate* exposure 4.5-6.0% air
exposed members not subjected to moisture saturation & chemicals
- *severe* exposure 5-7%

Mixture Design Procedures

Step 4: Estimate mixing
water

PCA Table 7-6

- Lb. Of water per yd^3
- Function of:
 - Slump
 - Air
 - Max Aggregate Size

Step 5: Estimate w/cm
ratio

PCA Tables 7-1,2,3

- Typically = 0.45

Table 7-6. Approximate Mixing Water and Target Air Content Requirements for Different Slumps and Maximum Sizes of Aggregate

Slump, in.	Water, pounds per cubic yard of concrete, for indicated maximum sizes of aggregate*							
	¾ in.	½ in.	¾ in.	1 in.	1½ in.	2 in.**	3 in.**	6 in.**
Non-air-entrained concrete								
1 to 2	350	335	315	300	275	260	220	190
3 to 4	355	355	340	325	300	285	245	210
6 to 7	410	385	360	340	315	300	270	—
Approximate amount of entrapped air in non-air-entrained concrete, per cent	3	2.5	2	1.5	1	0.5	0.3	0.2
Air-entrained concrete								
1 to 2	305	295	280	270	250	240	205	180
3 to 4	340	325	305	295	275	265	225	200
6 to 7	365	345	325	310	290	280	260	—
Recommended average total air content, per cent, for level of exposure:†								
Mild exposure	4.5	4.0	3.5	3.0	2.5	2.0	1.5	1.0
Moderate exposure	6.0	5.5	5.0	4.5	4.5	4.0	3.5	3.0
Severe exposure	7.5	7.0	6.0	6.0	5.5	5.0	4.5	4.0

*These quantities of mixing water are for use in computing cement factors for trial batches. They are maximums for reasonably well shaped angular coarse aggregates graded within limits of accepted specifications.

**The slump values for concrete containing aggregate larger than 1½ in. are based on slump tests made after removal of particles larger than 1½ in. by wet screening.

†The air content in job specifications should be specified to be delivered within -1 to +2 percentage points of the table target value for moderate and severe exposures.

Adapted from References 7-6 and 7-9.

Mixture Design Procedures

Step 6: Choose Cement Type

- Portland Cement
 - Types I-V
 - Generally type I or II
- Pozzolans
 - Fly Ash
 - Blast Furnace Slag
 - Silica Fume
 - * Decrease PC demand

Mixture Design Procedures

Step 7: Calculate the
cementitious content

$$\frac{\text{Water Content}}{w / \text{cm}} = \text{CM Content}$$

Mixture Design Procedures

Step 8: Estimate Coarse
Aggregate Content
PCA Table 7-5

- *Coarse Agg. Factor*
(CAF) = % Agg. in
concrete volume

Calculate the Coarse
Aggregate

- $CAF * DRUW * V_{concrete}$
- Affects workability

Step 9: Calculate the
Fine Aggregate

- $V_{concrete} - V_{constituents}$

Table 7-5. Volume of Coarse Aggregate Per Unit of Volume of Concrete

Maximum size of aggregate, in.	Volume of dry-rodded coarse aggregate* per unit volume of concrete for different fineness moduli of fine aggregate			
	2.40	2.60	2.80	3.00
3/8	0.50	0.48	0.46	0.44
1/2	0.59	0.57	0.55	0.53
3/4	0.66	0.64	0.62	0.60
1	0.71	0.69	0.67	0.65
1 1/2	0.75	0.73	0.71	0.69
2	0.78	0.76	0.74	0.72
3	0.82	0.80	0.78	0.76
6	0.87	0.85	0.83	0.81

*Bulk volumes are based on aggregates in dry-rodded condition as described in ASTM C 29. These volumes are selected from empirical relationships to produce concrete with a degree of workability suitable for usual reinforced construction. For less workable concrete such as required for concrete pavement construction, they may be increased about 10%. For more workable concrete, such as may sometimes be required when placement is to be by pumping, they may be reduced up to 10%.

Adapted from Reference 7-6.

Mixture Design Procedures

Step 10: Admixtures

- Air entraining agent
- water reducer
- ? accelerator
- ? retarder
- ? other

Mixture Design Procedures

- PCA Procedure is widely applicable
- No first trial is perfect
- Initial trial batch.....
- Determine
 - slump, air content
 - strength
- Adjust mix design
- Repeat as necessary